NAYAB KHUSHK

Manual QA Engineer | AI Solutions Builder
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SUMMARY

Computer Systems Engineering graduate with hands-on experience in manual software testing and AI-assisted application development. Tested 2 CRM systems, identifying 15+ critical defects, and built and evaluated a RAG chatbot over 60,000+ rows of data. Skilled in edge-case testing, prompt engineering, and practical LLM application skills.

EDUCATION

BE Computer Systems Engineering | Mehran University of Engineering & Technology Nov 2021 – Nov 2025

KEY SKILLS

Manual QA & Testing: Functional, Regression, Smoke, Sanity, End-to-End, UAT, Exploratory, Ad-Hoc, Test Case Design, Defect Lifecycle, Bug Reporting, RTM, RBAC, SDLC/STLC, Agile/Scrum

AI/LLM Testing: Hallucination, Response Accuracy & Latency, Prompt Regression, Boundary-Condition Testing, AI Feature Bug Reporting

Tools: JIRA, Postman, Git/GitHub, Browser DevTools, Google Lighthouse, Google Sheets, Replit, Google Antigravity, Google Colab

AI Solutions: Python (AI-Assisted), Prompt Engineering, RAG, LLM APIs (Groq, OpenAI Whisper), Hugging Face, Sentence Transformers, TensorFlow/Keras

WORK EXPERIENCE

Artificial Intelligence Intern | YOC – YourOnlineConversation®, Pakistan Jan 2026 – Apr 2026

• Defined and executed functional/regression testing for 2 CRM systems, identifying 15+ critical defects involving revenue calculations, duplicate booking logs, and RBAC overrides.

• Directed AI-assisted development of a domain-specific RAG chatbot over 27 JSON files and a 60,000+ row variation CSV, using semantic guardrails and cosine-similarity vector search (all-MiniLM-L6-v2), reducing hallucinations by 35% across 40+ benchmark queries.

• Tested LLM latency, response accuracy, and boundary conditions, documented defects and discussed reproducible issues with developers.

Software Testing Intern | GCS (Pvt.) Ltd. Jun 2025 – Jul 2025

• Tracked 20+ defects in JIRA and verified fixes with developers, reducing reopened tickets by 25% at sprint close.

• Performed post-deployment smoke/sanity testing on a Call Management System and identified UI alignment defects in an AI Based Attendance Management System.

Web Developer Intern | Pakistan Engineers Forum (PEF) – MUET Apr 2025 – May 2025

• Planned and evaluated a responsive HTML5/CSS3 prototype across devices using DevTools, improved Google Lighthouse performance from 70% to 90% through image compression and lazy loading.

PROJECTS

AI QA & Bug Investigation Agent (In Progress) – Python, FAISS, BM25, Sentence-Transformers, Cross-Encoder, pytest

• Designed an evidence-grounded RAG system that investigates bug reports against requirements, API docs, test cases, and past bugs from a 31-document CRM knowledge base, with rules to never invent IDs and to flag insufficient evidence.

• Built hybrid retrieval (FAISS semantic search, BM25 keyword search, metadata filtering, query caching) that returns results in 44 ms per query.

• Added cross-encoder re-ranking (ms-marco-MiniLM-L-6-v2): on the query "Login fails after password reset," duplicate chunks in the top 5 fell from 3 to 2, unique source documents rose from 3 to 4, and test case TC-AUTH-014 surfaced.

• Tracked retrieval and reranking scores and latency (1.2 s full pipeline on CPU, 10 candidates scored) and verified the re-ranker with 6 passing pytest tests (relevance ordering, metadata, diversity filter, score cutoff, latency metrics, empty inputs).

Non-Invasive Blood Pressure Prediction using Conv-LSTM (FYP) – Python, TensorFlow, Keras, OpenCV

• Trained a 1D CNN (3 convolution blocks, 64/32/16 filters) on the team's facial-video dataset, reaching a mean error of −3.1 mmHg (systolic) and −2.0 mmHg (diastolic), within the AAMI ±5 mmHg limit, and an MAE of 9.25 / 7.25 mmHg against Omron BP monitor readings.

• Benchmarked the CNN against the team's Conv-LSTM and ResNet models (70/10/20 train/validation/test split, Adam optimizer); Conv-LSTM performed best (MAE 8.62 / 6.38 mmHg) and was integrated by the team into a Flutter mobile app through a Django REST API.

SkillTek Web Portal – QA Bug Audit & Defect Documentation

• Audited the Admin Portal, Teacher Portal, and Public Website, and documented 8 critical defects, including assignment creation with 0 marks, future-dated attendance, duplicate student CNIC enrollments, and record deletion without confirmation.

CERTIFICATIONS

Project Management – Saylor University | Use AI as a Creative & Expert Partner – Google/Coursera | Generative AI – HEC x NCEAC x iCodeGuru | Packet Tracer & Networking Basics – Cisco